

PROJECT DOCUMENTATION

ChatSphere

MERN Real-Time Chat Application

A full-stack chat project I built with MongoDB, Express.js, React, Node.js, JWT authentication, Socket.IO real-time updates, and Cloudinary image uploads.

Stack

MERN, Socket.IO, MUI

Focus

Realtime chat and auth

Docs

Architecture and UI walkthrough

Prepared as project documentation for the ChatSphere repository.

Overview

I built ChatSphere as a practical MERN chat application, not just as a static UI. The project includes account creation, secure login, protected API routes, a real-time messaging dashboard, user presence, image support, editable messages, soft-deleted messages, and a profile settings page.

The backend is responsible for security, persistence, file upload integration, and socket events. The frontend is responsible for the user journey, API calls, socket connection, message state, profile updates, and the visual layout.

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Technology Stack

MongoDB

Mongoose

Express.js

Node.js

React

Vite

Material UI

Socket.IO

JWT

bcryptjs

Cloudinary

Multer

Axios

Frontend

I used React with Vite because it keeps development fast and simple. Material UI handles the main interface components, layout, buttons, forms, avatars, badges, and theme setup. Axios is used through one shared HTTP client so all API calls use the same base URL and cookie settings.

Backend

I used Express.js with a clear route-controller-model structure. MongoDB stores users and messages through Mongoose models. JWT cookies protect private routes, bcryptjs hashes passwords, and Cloudinary stores uploaded images.

Repository Structure

I kept the project split into backend, frontend, and documentation folders. This makes the repo easier to scan and avoids mixing server code, UI code, and project documents.

```
mern-chat-app-jwt-socketio/  
|-- backend/  
|   |-- src/  
|   |   |-- config/  
|   |   |-- controllers/  
|   |   |-- middleware/  
|   |   |-- models/  
|   |   |-- routes/  
|   |   |-- utils/  
|   |   `-- index.js  
|   |-- .env.example  
|   `-- package.json  
|-- frontend/  
|   |-- ui/  
|   |   |-- src/  
|   |   |   |-- api/  
|   |   |   |-- app/  
|   |   |   |-- constants/  
|   |   |   |-- pages/  
|   |   |   `-- theme/  
|   |   |-- .env.example  
|   |   `-- package.json  
|-- docs/  
|   |-- screenshots/  
|   |-- project-documentation.html  
|   `-- project-documentation.pdf  
|-- .gitignore  
|-- package.json  
`-- README.md
```

The root `package.json` works like a small command center. It lets me run backend and frontend commands from the main project folder while still keeping each app's dependencies inside its own folder.

Backend Implementation

Authentication

Signup validates the required fields, checks duplicate emails, hashes the password, saves the user, and creates a JWT cookie. Login compares the password with the stored hash and refreshes the cookie. Logout clears the cookie.

Protected Routes

The auth middleware reads the JWT from the cookie, verifies it, loads the user from MongoDB, removes the password field, and attaches the user to the request before allowing private routes to continue.

Messages

Message APIs load conversation history, create messages, edit own messages, and soft-delete own messages. Soft delete keeps the conversation timeline stable while hiding the original text and image from the UI.

Uploads

Profile image uploads use Multer memory storage and Cloudinary upload streams. Chat image messages are sent as Base64 from the client and then uploaded to Cloudinary before the message is saved.

Real-Time Flow

The main chat experience combines REST APIs with Socket.IO. I used REST for the main write operations because it keeps validation, authentication, and database saving in the same backend flow. After the database change is complete, Socket.IO broadcasts the update to the sender and receiver.

1. Auth Check

The frontend checks the logged-in user through the protected auth endpoint.

2. Socket Connect

The client connects to Socket.IO with the current user id in the query.

3. Save Message

The backend saves the message in MongoDB through the protected message API.

4. Broadcast

The backend emits events so both chat participants update instantly.

Socket Events I Used

- `getOnlineUsers` keeps the sidebar online/offline state updated.
- `newMessage` appends new messages to the active conversation.
- `messageUpdated` replaces the edited message in the current UI state.
- `messageDeleted` marks a message as deleted without removing its record.

The backend also has a direct socket `sendMessage` handler. The current UI path mainly uses the REST endpoint for sending messages, then relies on Socket.IO for the live update. That gives the project a clean persistence-first flow.

Frontend Implementation

Auth Screens

The signup and login pages use controlled React form state, simple validation, API error handling, and navigation after successful requests. The interface uses a dark, focused layout instead of a plain default form.

Dashboard

The dashboard has a compact icon rail, user list, online badges, message panel, emoji picker, image picker, edit controls, delete controls, and a send box. It loads users and conversation history from the backend.

Profile

The profile page lets the user update display name, preview a selected image, upload the profile picture, choose a theme, return to chat, or logout.

API Layer

I separated API calls into small files under `src/api`. The shared Axios client uses `withCredentials`, so the JWT cookie is included automatically when calling protected endpoints.

Security and Data Handling

- Passwords are hashed with bcryptjs before saving to MongoDB.
- JWT is stored in an HTTP-only cookie instead of localStorage.
- Private API routes use middleware to verify the cookie before continuing.
- CORS is configured with allowed client origins and credentials support.
- Profile image upload uses server-side Cloudinary upload handling.
- Deleted messages are soft-deleted, so the chat history does not break.

Local Setup

```
npm run install:all  
npm run dev:backend  
npm run dev:frontend
```

The backend runs on `http://localhost:3000` by default. The Vite frontend normally runs on `http://localhost:5173`. MongoDB and Cloudinary credentials should be added to `backend/.env` using `backend/.env.example` as the template.

What I Would Improve Next

- Add unread message counts and notification badges.
- Support multiple active sockets per user more explicitly.
- Add typing indicators and delivered/read status.
- Add backend tests for auth and message permissions.
- Add deployment configuration for frontend and backend hosting.

This document is written to explain the project as an individual full-stack build, including the implementation decisions and the current feature set.